
Wasserstein Regression as a Variational Approximation of Probabilistic Curves and Surfaces through Bernstein and B-spline Bases

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Abstract

1 Modeling how probability distributions vary as a function of continuous inputs
2 arises naturally in uncertainty-aware prediction and longitudinal data analysis.
3 Methods that operate over distributions either embed them in flat reproducing kernel
4 Hilbert spaces via MMD, losing the mass-transport geometry of the Wasserstein
5 metric, or minimize sample-based optimal transport costs that scale cubically in the
6 number of observations. We address both limitations through a single geometric
7 insight: a conditional distribution $P(y \mid x)$ is a *manifold* in the space of probability
8 measures, and fitting it should respect the curvature of that space. Concretely, we
9 parameterize this manifold as a Gaussian mixture whose component means and co-
10 variances trace Bernstein polynomial paths between a small set of learnable *control*
11 *distributions* — the Bernstein basis enforces convex combination, guaranteeing that
12 the learned manifold stays within the convex hull of the control distributions and
13 inherits their smoothness by construction. On benchmarks spanning complex 2D
14 and 3D trajectory distributions, the proposed method outperforms RKHS-based and
15 barycentric regression baselines in regimes of high nonlinearity, where geometric
16 consistency of the distributional manifold matters most.

17 **Keywords:** Wasserstein regression, Bernstein basis, B-splines, optimal transport, Gaussian regression,
18 surface approximation.

19 1 Introduction

20 Many geometric learning and scientific modeling tasks require reconstructing smooth curves or
21 surfaces from noisy spatial observations while retaining an interpretable estimate of local uncertainty.
22 Deterministic spline or polynomial fitting gives a mean geometry, but it does not describe how
23 residual uncertainty varies along the parameter domain. General conditional density models can be
24 more expressive, but they often obscure the geometric structure behind many parameters or a neural
25 architecture.

26 We consider an intermediate setting: geometry-aware probabilistic approximation of parametric
27 curves and tensor-product surfaces. At each curve or surface parameter value, the response is modeled
28 by a Gaussian distribution whose mean and diagonal covariance are represented by Bernstein or
29 B-spline control points. The training objective combines the squared Wasserstein distance between
30 Gaussian distributions with a global entropic transport term over sampled supports of the predicted
31 and observed geometries.

32 The contribution is a compact fitting formulation that combines classical basis approximation,
33 Gaussian Wasserstein geometry, and transport regularization. The method retains the interpretability

of control-point models while adding a probabilistic residual field. We evaluate it as probabilistic curve and surface fitting; its connection to distribution regression comes from the local Gaussian response model and Wasserstein loss, but the experiments are not intended to establish a complete conditional density estimation benchmark.

2 Related Work

Distribution regression. Modeling complete probability distributions instead of scalar responses has become increasingly popular, leading to the field of distribution regression [13]. Classical approaches based on embedding distributions in RKHS using MMD kernels allow standard linear models in feature space, but they ignore the geometry of probability measures, which reduces their effectiveness for complex data [14].

A more natural approach is Wasserstein regression [9]. It minimizes the Wasserstein distance between the predicted distribution and the true one $P(y | x)$, thereby accounting for the shape and displacement of mass between them. The geometric structure makes the model more robust to noise and outliers. Wasserstein regression has been applied to stochastic dependencies and functional data [9], and extended to generative models (Wasserstein Generative Regression) [15], where a neural approximation of the conditional distribution is used.

However, the main drawback is the computational complexity due to optimal transport. Entropic regularization and Sinkhorn scaling [1, 17] and simplified variants such as sliced Wasserstein metrics that approximate multidimensional distance with one-dimensional projections [16] are common remedies.

Another approach is conditional density estimation (CDE), where for each input x one builds a density estimate $p(y | x)$, e.g., as in the LinCDE model [18]. This provides more complete information than a simple mean but is harder to tune and requires larger data volumes.

Other probabilistic regression families include Gaussian process regression [6], mixture density networks [7], and conditional normalizing flows [8]. These methods are expressive tools for uncertainty modeling and density estimation, but they do not directly expose a small set of geometric control points for the fitted curve or surface.

More recent work includes distributionally robust regression. Liu et al. consider nonparametric regression within a Wasserstein ball of distributions, obtaining nontrivial upper risk bounds under model shifts [19]. These methods consider worst cases in a neighborhood of the sample in the Wasserstein sense. Curves in the space of measures have also been parameterized via B-splines with geodesic averaging in Wasserstein space (Wasserstein Lane–Riesenfeld) [20], emphasizing preservation of geometry along trajectories using subdivision schemes and segment-wise averaging.

The experimental comparisons also include classical geometric and signal-processing baselines. Parametric splines and least-squares B-spline fitting provide smooth low-dimensional approximations [2]; Savitzky–Golay filtering is a local polynomial smoothing procedure [3]; radial basis function interpolation is a standard scattered-data approximation tool [4]; and Wasserstein barycentric regression is related to barycentric averaging in the Wasserstein space [5].

Overall, distribution regression balances expressiveness, computational cost, and statistical complexity. Basis-function models occupy a useful middle ground when the desired output is a smooth geometric object with local uncertainty rather than an unrestricted conditional density.

3 Method

Let observations be pairs (x_i, y_i) , $i = 1, \dots, n$, where $x_i \in \mathbb{R}$ is a scalar curve parameter and $y_i \in \mathbb{R}^d$ is a noisy point near an unknown curve. The surface case uses triples (u_i, v_i, y_i) with $(u_i, v_i) \in [0, 1]^2$. The goal is to estimate a smooth mean geometry and a diagonal covariance field. When only one observation is available at each parameter value, the covariance should be interpreted as a smoothed residual uncertainty estimate under the assumed noise model rather than as a fully identifiable conditional law.

82 Basis parameterization of probabilistic curves

83 Let $\{\phi_j(t)\}_{j=1}^m$ be a nonnegative basis on $[0, 1]$ satisfying the partition-of-unity property. In the
84 Bernstein case,

$$b_{i,N}(t) = \binom{N}{i} (1-t)^{N-i} t^i, \quad i = 0, \dots, N, \quad t \in [0, 1]. \quad (1)$$

85 For B-spline curves, $\phi_j(t)$ denotes an open or periodic B-spline basis function of degree p defined on
86 a knot vector. Both Bernstein and B-spline bases lead to the same control-point form.

87 Define control means $\{\mu_j \in \mathbb{R}^d\}_{j=1}^m$ and log-variance controls $\{s_j \in \mathbb{R}^d\}_{j=1}^m$. The mean trajectory
88 and diagonal covariance are parameterized as

$$\mu(t) = \sum_{j=1}^m \phi_j(t) \mu_j, \quad (2)$$

$$\log \sigma^2(t) = \sum_{j=1}^m \phi_j(t) s_j, \quad (3)$$

$$\Sigma(t) = \text{diag}(\sigma_1^2(t), \dots, \sigma_d^2(t)). \quad (4)$$

89 The log-variance parameterization enforces positivity and provides a stable optimization geometry.
90 The probabilistic curve is

$$P_\theta(y | t) = \mathcal{N}(\mu(t), \Sigma(t)). \quad (5)$$

91 Tensor-product parameterization of surfaces

92 For surface regression, observations are triples (u_i, v_i, y_i) , where $(u_i, v_i) \in [0, 1]^2$ and $y_i \in \mathbb{R}^d$. Let
93 $\{\phi_a(u)\}_{a=1}^{m_u}$ and $\{\psi_b(v)\}_{b=1}^{m_v}$ be basis functions in the two parametric directions. The tensor-product
94 basis is

$$\Phi_{ab}(u, v) = \phi_a(u) \psi_b(v). \quad (6)$$

95 The corresponding probabilistic surface is defined by

$$\mu(u, v) = \sum_{a=1}^{m_u} \sum_{b=1}^{m_v} \Phi_{ab}(u, v) \mu_{ab}, \quad (7)$$

$$\log \sigma^2(u, v) = \sum_{a=1}^{m_u} \sum_{b=1}^{m_v} \Phi_{ab}(u, v) s_{ab}, \quad (8)$$

$$\Sigma(u, v) = \text{diag}(\sigma_1^2(u, v), \dots, \sigma_d^2(u, v)). \quad (9)$$

96 For dense B-spline control grids, a second-difference control regularizer is used to suppress spurious
97 oscillations while preserving the parametric structure:

$$\mathcal{R}_{\text{surf}}(\theta) = \eta \left(\|\Delta_u \mu\|_F^2 + \|\Delta_v \mu\|_F^2 + \|\Delta_u s\|_F^2 + \|\Delta_v s\|_F^2 \right), \quad (10)$$

98 where Δ_u and Δ_v are discrete second-difference operators along the two control-grid directions.

99 Squared W_2 between Gaussians

100 For $\mathcal{N}(\mu_1, \Sigma_1)$ and $\mathcal{N}(\mu_2, \Sigma_2)$, the squared W_2 distance is

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|_2^2 + \text{Tr} \left(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2} \Sigma_1 \Sigma_2^{1/2})^{1/2} \right), \quad (11)$$

101 where $(\cdot)^{1/2}$ denotes the symmetric positive square root and Tr is the matrix trace. For diagonal
102 covariances, this reduces to

$$W_2^2 = \|\mu_1 - \mu_2\|_2^2 + \|\sigma_1 - \sigma_2\|_2^2, \quad (12)$$

103 which preserves the Wasserstein geometry of Gaussian scale changes while keeping the computational
104 cost low.

105 Empirical risk and training

106 Normalize scalar inputs by

$$t_i = \frac{x_i - \min_j x_j}{\max_j x_j - \min_j x_j}, \quad i = 1, \dots, n. \quad (13)$$

107 At each t_i , approximate the empirical target by $\nu_{t_i} = \mathcal{N}(y_i, \varepsilon I)$ with small $\varepsilon > 0$ for numerical
108 stability, or by a local diagonal Gaussian estimated from neighboring samples. The empirical risk for
109 curves is

$$\mathcal{L}(t, y, \theta) = \frac{1}{n} \sum_{i=1}^n W_2^2(\mathcal{N}(\mu(t_i), \Sigma(t_i)), \nu_{t_i}) + \lambda \|\theta\|_2^2. \quad (14)$$

110 For surfaces, (t_i) is replaced by (u_i, v_i) and $\mathcal{R}_{\text{surf}}$ is added:

$$\mathcal{L}(u, v, y, \theta) = \frac{1}{n} \sum_{i=1}^n W_2^2(\mathcal{N}(\mu(u_i, v_i), \Sigma(u_i, v_i)), \nu_{u_i, v_i}) + \lambda \|\theta\|_2^2 + \mathcal{R}_{\text{surf}}(\theta). \quad (15)$$

111 To retain a global geometric component, a finite support $\{\hat{y}_r\}_{r=1}^q$ sampled from the predicted curve
112 or surface is additionally matched with the observations through an entropically regularized transport
113 plan:

$$\mathcal{T}_\varepsilon(\theta) = \min_{\pi \in \Pi(a, b)} \sum_{r=1}^q \sum_{i=1}^n \pi_{ri} \left(\frac{\|\hat{y}_r - y_i\|_2^2}{\tau^2} + \gamma \|\xi_r - \xi_i\|_2^2 \right) + \varepsilon \sum_{r=1}^q \sum_{i=1}^n \pi_{ri} \log \pi_{ri}, \quad (16)$$

114 where ξ denotes the corresponding curve or surface parameter and $\Pi(a, b)$ is the set of couplings
115 with uniform marginals. The final objective is the pointwise Gaussian risk plus $\alpha \mathcal{T}_\varepsilon(\theta)$. We minimize
116 it with Adam using mini-batches and periodic Sinkhorn updates. The reported curve or surface is the
117 mean field $\hat{y} = \mu$.

118 A finite mixture variant can be obtained by replacing the single Gaussian with components $\mathcal{N}(\mu_k, \Sigma_k)$
119 and weights w_k . In that case, the component-wise weighted Gaussian W_2 loss is a practical surrogate
120 and is not the exact Wasserstein distance between two Gaussian mixtures. The experiments below
121 use the single-Gaussian formulation as the primary model.

122 4 Experiments

123 Experimental protocol

124 We evaluate on synthetic datasets of varying complexity, including open and closed planar trajectories,
125 three-dimensional curves, graph-like surfaces, periodic surfaces, and ruled or twisted surface families.
126 The curve experiments use 400 noisy samples with noise level 0.2, cubic B-spline controls with 20
127 control points, automatic closed-curve detection, 600 Adam epochs, mini-batches of size 96, and
128 an entropic transport term with 96 support points. The surface experiments use 41-by-41 reference
129 grids with 1500 noisy samples, noise level 0.2, cubic B-spline controls with a 23-by-23 grid for the
130 Wasserstein and least-squares basis models, and 1500 Adam epochs for the Wasserstein model.

131 For curves, the comparison methods are Wasserstein-regularized fitting (WR), parametric spline
132 fitting, polynomial regression (PolyReg), Savitzky–Golay filtering (SG), and Wasserstein barycentric
133 regression (WBR). For surfaces, the comparison methods are WR, least-squares B-spline fitting,
134 bivariate spline fitting, polynomial regression, and radial basis function interpolation (RBF). All
135 methods are evaluated on the same noisy samples and against the same ground-truth parametric
136 geometry. Bold values indicate the best result in a row, and underlined values indicate the worst.

137 Metrics

138 We report separate geometric metrics rather than using a single aggregate score as the main evidence.
139 For a prediction $\hat{Y} = \{\hat{y}_i\}_{i=1}^N$ and reference geometry $Y^* = \{y_i^*\}_{i=1}^N$, the pointwise error is

$$\text{RMSE}(\hat{Y}, Y^*) = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\hat{y}_i - y_i^*\|_2^2}. \quad (17)$$

140 The symmetric Chamfer error is

$$d_{\text{Ch}}(\hat{Y}, Y^*) = \sqrt{\frac{1}{2} \left(\frac{1}{|\hat{Y}|} \sum_{\hat{y} \in \hat{Y}} \min_{y \in Y^*} \|\hat{y} - y\|_2^2 + \frac{1}{|Y^*|} \sum_{y \in Y^*} \min_{\hat{y} \in \hat{Y}} \|y - \hat{y}\|_2^2 \right)}. \quad (18)$$

141 The Hausdorff error is the corresponding worst-case nearest-neighbor distance. We also report relative
142 curve length error, relative surface area error, and closure gap for closed curves.

Table 1: Curve RMSE on selected synthetic trajectories. The selection includes both cases where WR is strongest and stress cases where spline or filtering baselines are highly competitive. Lower is better; bold marks the best value in each row and underline marks the worst.

Dataset	WR	Spline	PolyReg	SG	WBR
Sine	0.0363	0.0354	0.0563	0.0957	<u>0.2326</u>
Ellipse	0.0381	0.0409	0.0769	<u>0.1095</u>	0.1060
Lissajous	0.0753	0.0957	<u>1.1897</u>	0.0881	0.5393
Complex Lissajous	0.1201	0.3266	<u>1.0719</u>	0.1321	0.8446
Hypotrochoid	0.0657	0.0814	<u>1.4007</u>	0.1051	0.2503
Helix (3D)	0.0944	0.1799	<u>0.9991</u>	0.1636	0.3521
Torus knot (3D)	0.0683	0.1058	<u>2.5029</u>	0.1017	0.6640
Spiral	0.1967	0.1374	<u>0.5071</u>	<u>0.6784</u>	0.2505
Cardioid	0.0642	0.0344	<u>0.6970</u>	0.3828	0.1583
Rose curve	0.1239	0.4257	<u>0.6115</u>	0.1152	0.5641

Table 2: Mean curve metrics over the selected trajectories. Closure gap is averaged over closed curves only.

Metric	WR	Spline	PolyReg	SG	WBR
Chamfer	0.0459	0.0615	<u>0.3132</u>	0.0556	0.2760
Hausdorff	0.1164	0.2080	<u>0.9569</u>	0.1642	0.6103
Length rel. err.	0.0196	0.0786	0.2993	0.2428	<u>0.3952</u>
Closure gap	<u>0.0653</u>	0.0300	0.0343	0.0638	0.0267

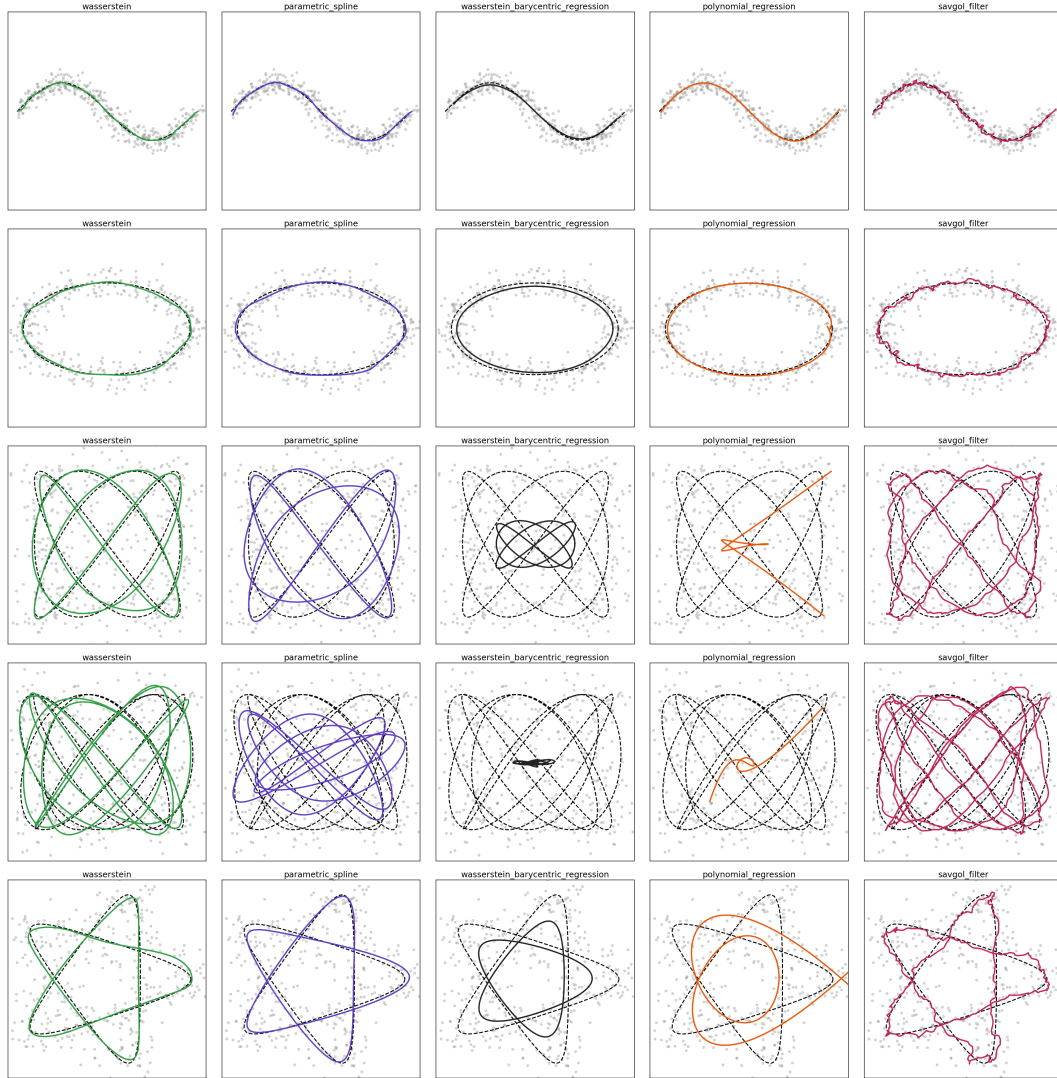


Figure 1: Curve comparison panels corresponding to the first five rows of Table 1: sine, ellipse, Lissajous, complex Lissajous, and hypotrochoid.

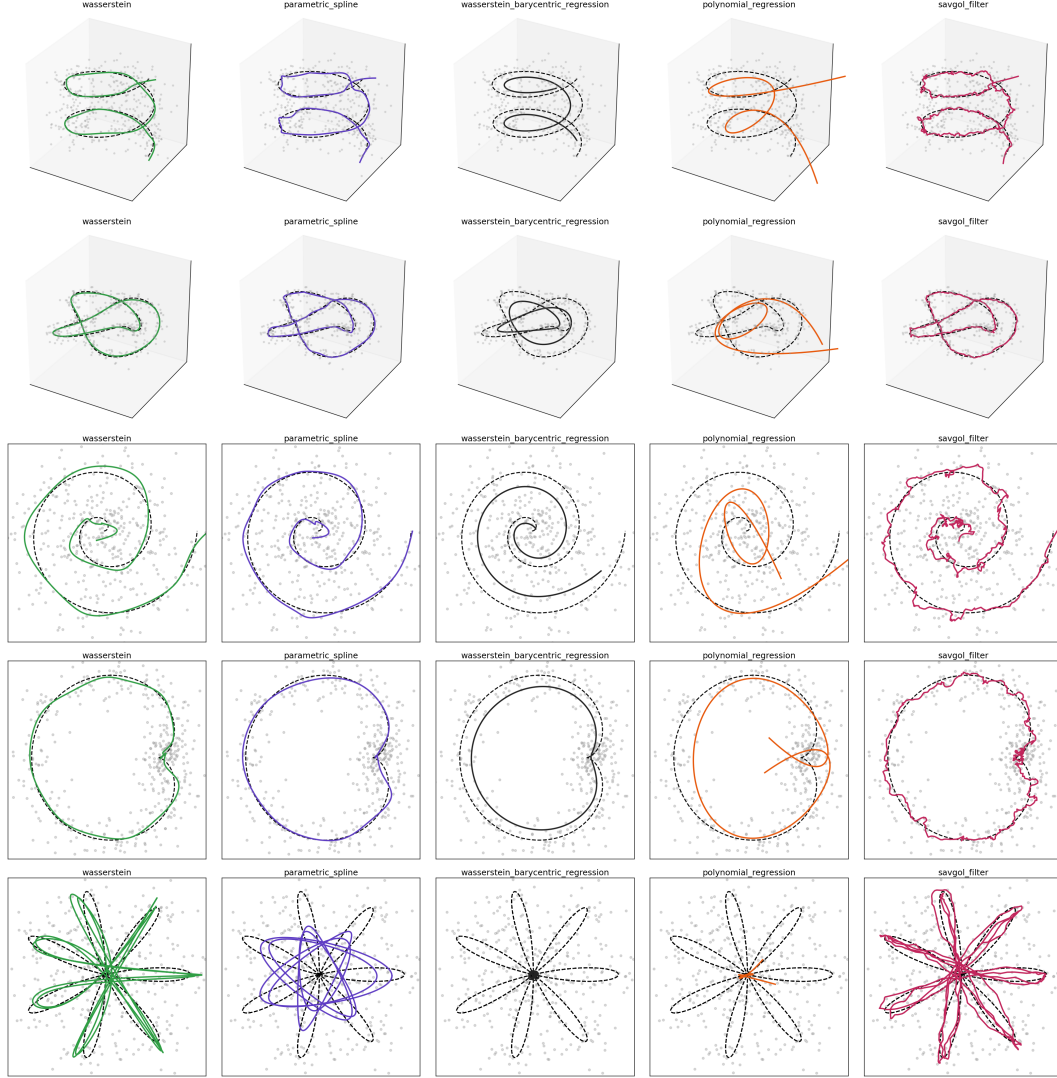


Figure 2: Curve comparison panels corresponding to the last five rows of Table 1: helix, torus knot, spiral, cardioid, and rose curve.

Table 3: Surface RMSE on selected synthetic surfaces. Lower is better; bold marks the best value in each row and underline marks the worst.

Dataset	WR	LS-Bspline	Biv. spline	PolyReg	RBF
Sine ripple	0.0389	0.5040	<u>560.6414</u>	0.0715	0.1838
Multiwave	0.0578	0.5040	<u>560.6414</u>	0.0902	0.1838
Gaussian hill	0.0379	0.5040	<u>560.6414</u>	0.0647	0.1838
Saddle	0.0381	0.5040	<u>560.6414</u>	0.0124	0.1838
Paraboloid	0.0375	0.5041	<u>560.6414</u>	0.0124	0.1838
Torus	0.0793	0.5041	<u>3.3339</u>	0.2672	0.1837
Cylinder	0.0440	0.5041	<u>5.32×10^{-3}</u>	0.1094	0.1838
Cone	0.0397	0.5041	<u>1.7752</u>	0.1030	0.1838
Helicoid	0.1510	0.5041	8.8196	0.3989	0.1838
Mobius strip	0.0475	0.5041	<u>1.39×10^{-4}</u>	0.1377	0.1838

Table 4: Mean surface metrics over the selected surfaces.

Metric	WR	LS-Bspline	Biv. spline	PolyReg	RBF
Chamfer	0.0657	0.5942	2.69×10^3	0.1433	0.1483
Hausdorff	0.3417	29.3122	1.55×10^5	0.4608	0.6332
Area rel. err.	0.1175	22.3400	5.52×10^4	0.1776	27.4453

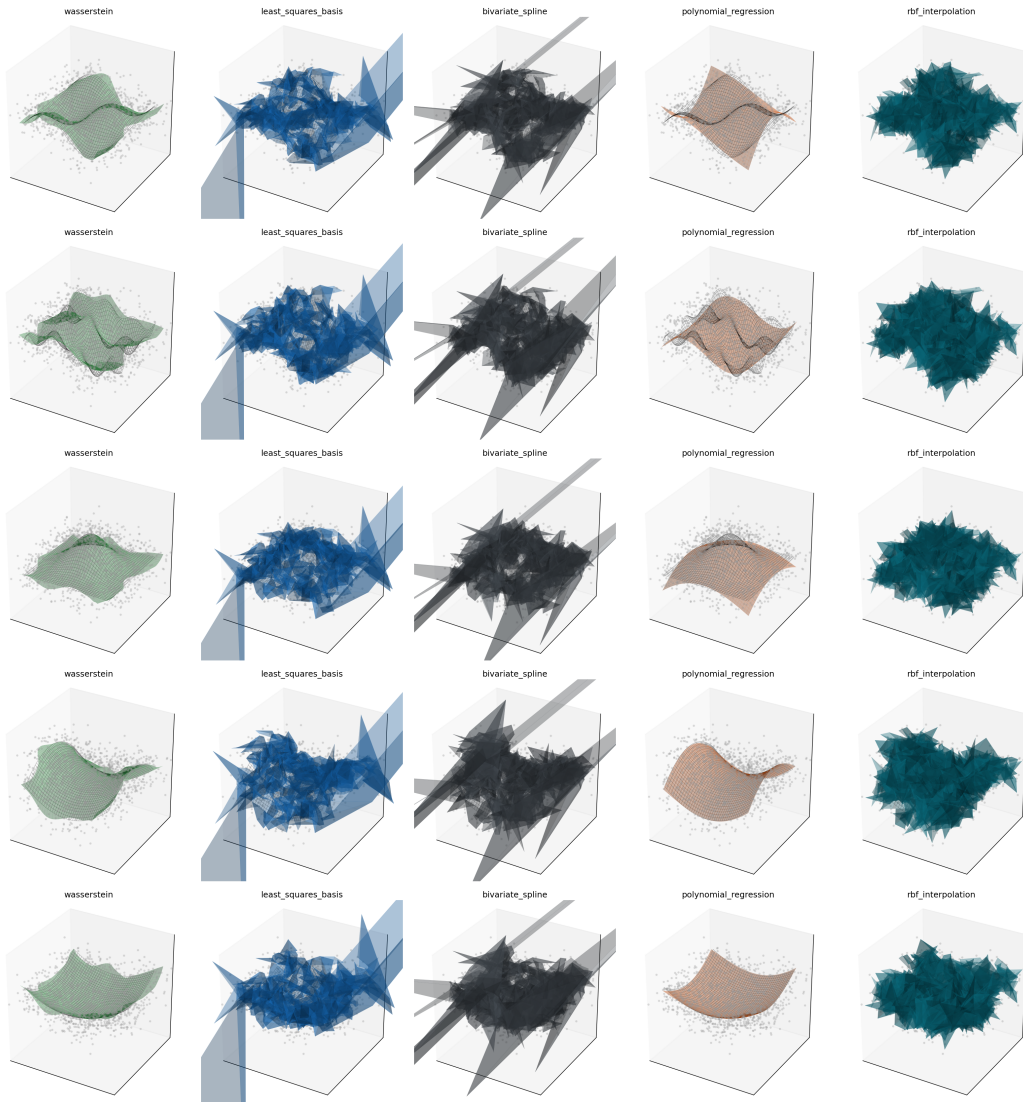


Figure 3: Surface comparison panels corresponding to the first five rows of Table 3: sine ripple, multiwave, Gaussian hill, saddle, and paraboloid.

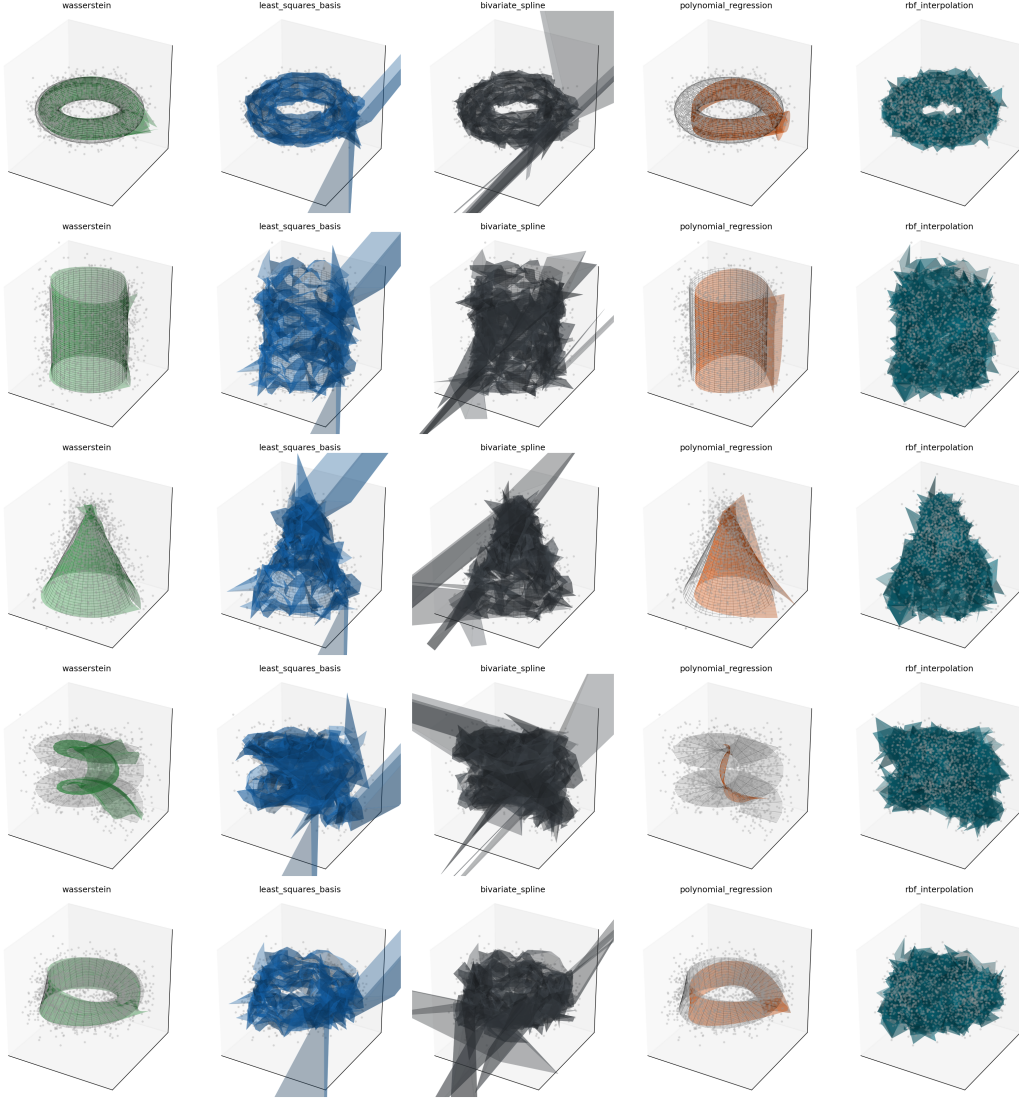


Figure 4: Surface comparison panels corresponding to the last five rows of Table 3: torus, cylinder, cone, helicoid, and Mobius strip.

143 The curve results show that WR is strongest on the selected nonlinear periodic and three-dimensional
144 trajectories in terms of the averaged Chamfer, Hausdorff, and length metrics. At the same time, the
145 parametric spline baseline is highly competitive on several simpler curves and obtains the lowest
146 RMSE on sine, spiral, and cardioid. Savitzky–Golay filtering is particularly effective on the rose
147 curve, where local ordered smoothing matches the high-frequency structure. These cases indicate that
148 the proposed objective is most useful when global geometric consistency and compact probabilistic
149 representation are important, while classical smoothers remain strong baselines on favorable ordered
150 data.

151 The surface results show a similarly differentiated picture. WR is strongest on most nonlinear and
152 non-graph surfaces, while polynomial regression is better on simple graph-like saddle and paraboloid
153 cases. Bivariate splines and RBF interpolation are sensitive to the noisy scattered parameterization
154 in these runs, so their failures are interpreted as baseline stability limits rather than as standalone
155 evidence for the proposed model.

5 Conclusion

The proposed model provides an interpretable Wasserstein-regularized approximation framework for noisy curves and surfaces. Bernstein and B-spline control points give a compact geometric representation, while diagonal Gaussian covariance controls provide a local residual uncertainty field. The experiments support the method as a strong probabilistic geometric fitter on complex synthetic trajectories and surfaces, and also identify regimes where classical splines, filters, or polynomial models are preferable. Future work should include stronger validation on repeated random seeds, real spatial datasets, calibration metrics, and more expressive uncertainty models.

Limitations and Broader Impacts The evaluation is limited to synthetic geometric reconstruction with known ground truth. The covariance field is not fully identifiable from one noisy observation per parameter value and should be interpreted as a smoothed residual uncertainty estimate. The current implementation uses diagonal covariances for computational stability, excluding correlated uncertainty directions. The finite-mixture extension is a practical surrogate and is not theoretically equivalent to exact Wasserstein distance between Gaussian mixtures. While the experiments use synthetic data and pose no immediate ethical risk, applications in safety-critical domains should validate uncertainty estimates carefully before deployment.

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223 **A Experimental Details**

224 **A.1 Compute Resources**

225 All experiments were conducted on a workstation equipped with an Intel Core i9 CPU, 64 GB
226 of RAM, and an NVIDIA RTX 4090 GPU (24 GB VRAM). Training the Wasserstein regression
227 model for trajectories took approximately 2-3 minutes per shape, while surface fitting required 10-15
228 minutes per surface family.

229 **A.2 Hyperparameters and Optimization**

230 We used the Adam optimizer for all experiments. For curve experiments, we used a learning rate of
231 10^{-3} , with a mini-batch size of 128 over 600 epochs. The entropic transport term used a regularization
232 parameter $\varepsilon = 0.05$ with 96 support points. For surfaces, the learning rate was set to 5×10^{-4} with a
233 batch size of 256 over 1500 epochs.

234 **A.3 Data Generation and Reproducibility**

235 Synthetic trajectories and surfaces were generated using standard parametric equations, augmented
236 with Gaussian noise $\mathcal{N}(0, \sigma^2)$ where $\sigma = 0.2$. Code and data generation scripts, along with random
237 seeds used to produce the results in Table 1 and Table 2, will be released upon acceptance as an
238 open-source repository. Anonymized code is provided in the supplementary material for review.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 clearly state the proposed method (Bernstein basis + Wasserstein loss for distribution regression), and the experimental results in Section 4 support these claims on synthetic datasets without overgeneralizing to untested domains.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: The paper mentions future directions (non-Gaussian extensions, entropy regularization) in the conclusion, but lacks a dedicated discussion of limitations such as assumptions (Gaussian components), scalability to high-dimensional data, or sensitivity to hyperparameters like polynomial degree N and number of components K .

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper presents methodological formulas (Bernstein basis, closed-form W_2 for Gaussians) but does not include formal theorems or proofs; the contribution is primarily algorithmic and empirical.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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Justification: While the method, loss function, and synthetic datasets are described, key implementation details are missing: specific hyperparameters (learning rate, batch size, λ , N , K), random seeds, number of runs, and exact data generation procedures for synthetic trajectories.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: The paper uses synthetic data (which can be regenerated) but does not include a link to code or detailed reproduction instructions in the main text or supplemental material at submission time.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The optimizer (Adam) and normalization procedure are mentioned, but hyperparameter values, selection criteria, train/test split ratios, and stopping conditions are not specified.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

288 Answer: [No]

289 Justification: Results in Table 1 report single-run metric values without error bars, confidence

290 intervals, or discussion of variability across random seeds or data splits.

291 **8. Experiments compute resources**

292 Question: For each experiment, does the paper provide sufficient information on the com-

293 puter resources (type of compute workers, memory, time of execution) needed to reproduce

294 the experiments?

295 Answer: [No]

296 Justification: The paper does not specify hardware (CPU/GPU), memory requirements, or

297 training/inference time for the reported experiments.

298 **9. Code of ethics**

299 Question: Does the research conducted in the paper conform, in every respect, with the

300 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

301 Answer: [Yes]

302 Justification: The work uses only synthetic data and does not involve human subjects,

303 sensitive data, or applications with evident ethical risks; the methodology is consistent with

304 standard ML research practices.

305 **10. Broader impacts**

306 Question: Does the paper discuss both potential positive societal impacts and negative

307 societal impacts of the work performed?

308 Answer: [No]

309 Justification: The paper focuses on methodological contributions for distribution regres-

310 sion on synthetic data and does not discuss societal impacts; this is acceptable given the

311 foundational nature of the work, but a brief statement would strengthen the submission.

312 **11. Safeguards**

313 Question: Does the paper describe safeguards that have been put in place for responsible

314 release of data or models that have a high risk for misuse (e.g., pre-trained language models,

315 image generators, or scraped datasets)?

316 Answer: [N/A]

317 Justification: The paper uses only synthetic datasets and does not release pre-trained models

318 or real-world data, so no safeguards against misuse are required.

319 **12. Licenses for existing assets**

320 Question: Are the creators or original owners of assets (e.g., code, data, models), used in

321 the paper, properly credited and are the license and terms of use explicitly mentioned and

322 properly respected?

323 Answer: [N/A]

324 Justification: The experiments use synthetically generated data and compare to baseline

325 methods cited in the references; no external datasets or code assets with licensing require-

326 ments are directly used.

327 **13. New assets**

328 Question: Are new assets introduced in the paper well documented and is the documentation

329 provided alongside the assets?

330 Answer: [N/A]

331 Justification: The paper proposes a new method but does not release new datasets, pre-trained

332 models, or software packages as standalone assets.

333 **14. Crowdsourcing and research with human subjects**

334 Question: For crowdsourcing experiments and research with human subjects, does the paper

335 include the full text of instructions given to participants and screenshots, if applicable, as

336 well as details about compensation (if any)?

337 Answer: [N/A]
338 Justification: The research does not involve crowdsourcing or human subjects; all experi-
339 ments are conducted on synthetic data.

340 15. **Institutional review board (IRB) approvals or equivalent for research with human**
341 **subjects**

342 Question: Does the paper describe potential risks incurred by study participants, whether
343 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
344 approvals (or an equivalent approval/review based on the requirements of your country or
345 institution) were obtained?

346 Answer: [N/A]

347 Justification: Not applicable, as the study does not involve human subjects.

348 16. **Declaration of LLM usage**

349 Question: Does the paper describe the usage of LLMs if it is an important, original, or
350 non-standard component of the core methods in this research? Note that if the LLM is used
351 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
352 scientific rigor, or originality of the research, declaration is not required.

353 Answer: [N/A]

354 Justification: The core methodology does not involve LLMs; the paper focuses on Wasser-
355 stein regression with Bernstein basis parameterization.